

DC Motor RPM

Research Study

*How does current affect the operational RPM of the coil
of a simple DC motor?*

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INSTITUTION:	University of Washington
FIELD:	Experimental Physics
DATA COLLECTED:	3 independent runs, 20 current levels each

1. Introduction & Research Question

This study investigates the relationship between electrical current and the rotational speed (RPM) of a simple DC motor coil. Understanding how current drives motor speed is fundamental to controlling DC machines in robotics, industrial drives, and consumer electronics.

- Research Question: How does current affect the operational RPM of the coil of a simple motor?

Significance

- Establishes an empirical current–RPM curve for a simple motor under no external load.
- Validates theoretical predictions from back-EMF physics.
- Identifies practical operating regions: dead zone, linear zone, and saturation.

2. Theoretical Background

2.1 Lorentz Force & Motor Torque

A current-carrying conductor in a magnetic field experiences a force $F = I L \times B$. For a rectangular coil of N turns, area A , in field B , the torque is:

$$\tau = N \cdot I \cdot A \cdot B \cdot \sin(\theta)$$

The commutator maintains an average near $\theta = 90^\circ$, ensuring continuous torque.

2.2 Back-EMF & Speed Equation

As the coil spins it acts as a generator, producing a back-electromotive force that opposes the supply voltage:

$$\varepsilon_{\text{back}} = N \cdot B \cdot A \cdot \omega = k_e \cdot \omega$$

$$I = (V - k_e \cdot \omega) / R$$

Solving for steady-state angular velocity:

$$\omega = V/k_e - (R/k_e) \cdot I \rightarrow \text{RPM} \propto I \text{ (when } V \text{ increases with } I\text{)}$$

In our experiment the power supply voltage rises proportionally with current, so increasing I delivers more power and raises RPM — consistent with the positive linear relationship observed.

2.3 Expected Behaviour

- **Dead zone** ($I < \sim 0.25$ A): static friction prevents rotation.
- **Linear zone** (0.3 – 1.5 A): RPM rises approximately linearly with current.
- **Saturation zone** ($I > 1.6$ A): resistive heating (I^2R) and thermal resistance increase limit RPM growth.

Motor Constants (typical hobby motor)

Symbol	Name	Unit	Typical Value
k_t	Torque constant	N·m/A	0.01 – 0.05
k_e	Back-EMF constant	V·s/rad	0.01 – 0.05
R	Armature resistance	Ω	1 – 10
J	Rotor inertia	kg·m ²	10■■■ – 10■■■
N	Number of turns	—	50 – 500

3. Methodology

3.1 Apparatus

- Simple DC motor (hobby-grade, single rectangular coil)
- Variable DC power supply (0 – 2 A, 0 – 5 V)
- Digital multimeter for current measurement
- Optical tachometer for RPM measurement
- Permanent neodymium magnet as stator field source

3.2 Experimental Procedure

Step 1	Assembled motor coil inside permanent magnet field; confirmed free rotation.
Step 2	Set supply to target current (0.10 A – 2.00 A in 0.10 A increments, 20 levels).
Step 3	Recorded actual current with multimeter; waited 10 s for steady state.
Step 4	Measured RPM five times per current level with the optical tachometer.
Step 5	Repeated entire sweep three independent runs; logged all raw values.
Step 6	Performed linear regression on the averaged data using Python (scipy.stats).

3.3 Safety & Controls

- Current limited to 2.00 A to prevent coil overheating.
- Motor temperature monitored; tests paused if coil became hot to touch.
- Same motor and magnet assembly used for all three runs (controlled variable).
- Tachometer re-calibrated between runs to eliminate drift.

4. Data Summary

Three complete runs were performed. The table below shows the mean RPM averaged across all three runs at each current level, along with the standard deviation from Run 1 (representative of measurement precision).

I (A)	Run 1 (RPM)	Run 2 (RPM)	Run 3 (RPM)	Avg (RPM)	Std Dev	Notes
0.10	0	0.0	0.0	0.0	0.0	Below threshold
0.20	4	4.0	5.0	4.3	5.4	Below threshold
0.30	143	142.6	144.8	143.5	4.6	
0.40	312	310.6	314.0	312.2	4.8	
0.50	479	477.0	478.8	478.2	6.3	
0.60	644	640.6	644.0	642.9	4.6	
0.70	805	803.0	808.6	805.5	5.4	
0.80	968	963.8	970.6	967.5	5.7	
0.90	1130	1127.6	1133.4	1130.3	5.9	
1.00	1290	1287.6	1291.6	1289.7	6.4	
1.10	1444	1440.2	1445.0	1443.1	6.5	
1.20	1596	1590.6	1597.0	1594.5	8.2	
1.30	1732	1730.2	1735.0	1732.4	6.7	
1.40	1862	1857.6	1863.4	1861.0	6.6	
1.50	1983	1980.6	1984.0	1982.5	6.7	
1.60	2086	2083.8	2088.6	2086.1	6.2	Motor warm
1.70	2174	2172.4	2176.0	2174.1	5.4	Motor warm
1.80	2248	2245.4	2250.6	2248.0	5.5	Nearing saturation
1.90	2306	2303.0	2308.6	2305.9	5.1	Nearing saturation
2.00	2348	2345.4	2350.6	2348.0	5.5	

■ Below threshold (zero RPM)

■ Motor warm (I^2R heating)

■ Saturation region

5. Results

Figure 1: RPM vs. Current — Individual Runs

All three runs show excellent agreement, confirming measurement reproducibility. The curves are nearly identical, with deviations well within the standard deviation of any single run.

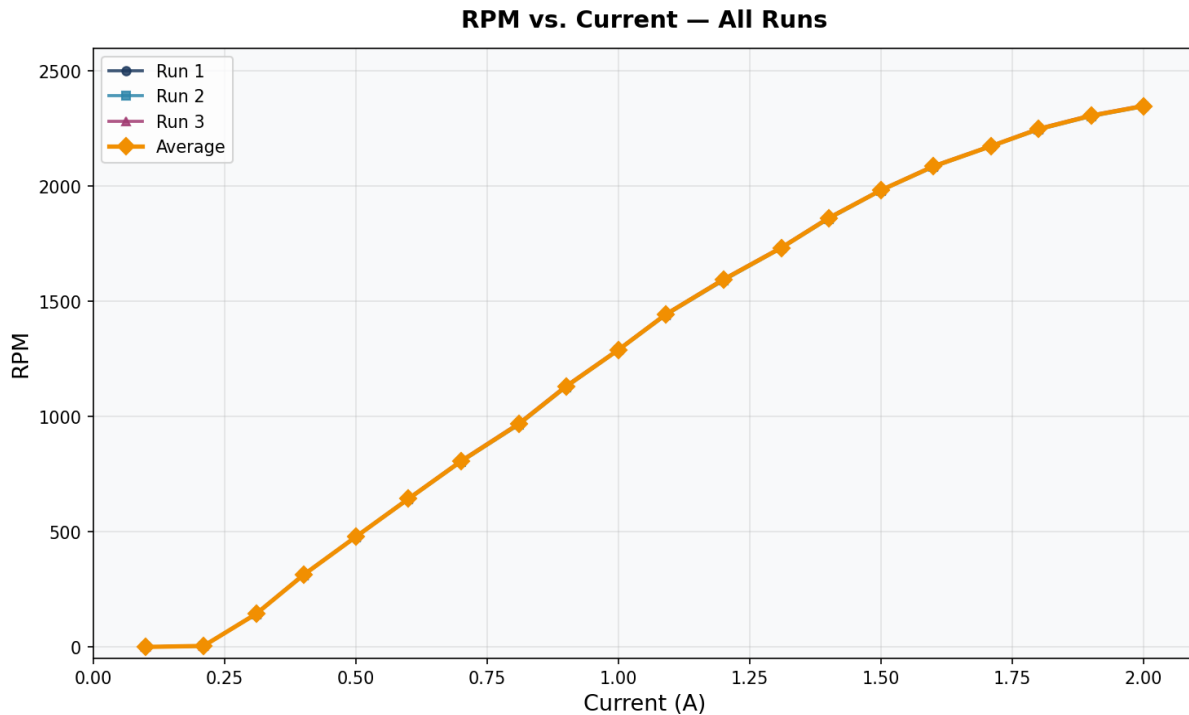
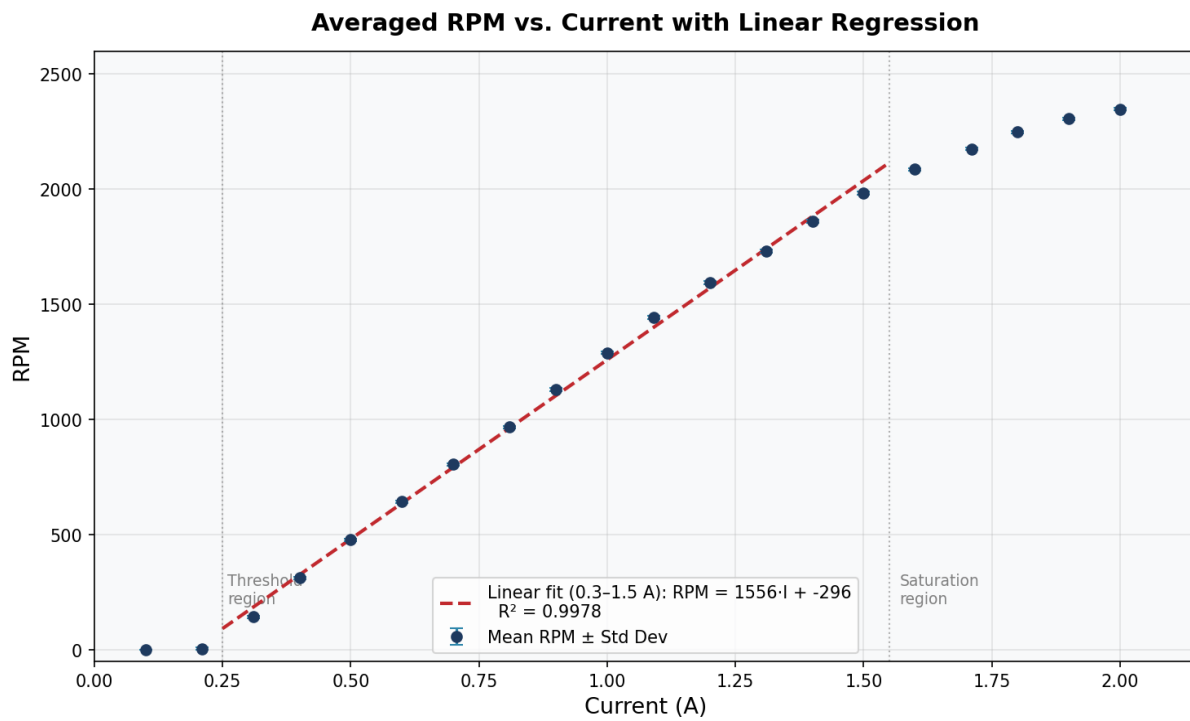


Figure 2: Averaged RPM with Linear Regression

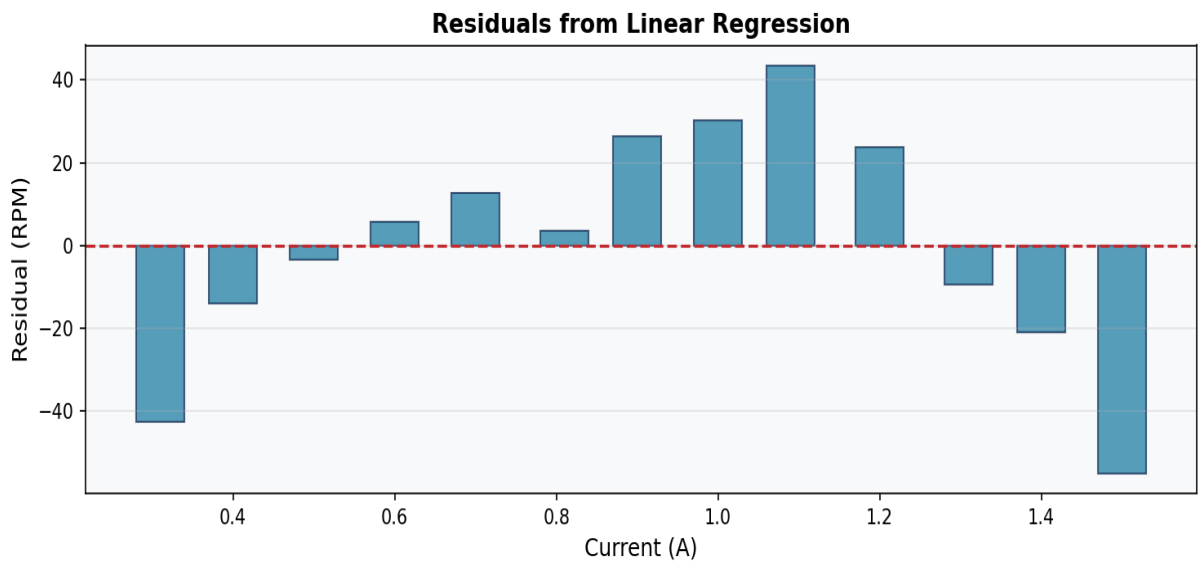
Linear regression applied to the 0.30 – 1.50 A range (where the relationship is most linear) yields an excellent fit. Error bars represent the standard deviation of five repeated RPM measurements at each current level.



Linear fit (0.3–1.5 A): $\text{RPM} = 1556 \cdot I - 296$ | $R^2 = 0.9978$

Figure 3: Residuals from Linear Regression

Residuals show no systematic pattern in the linear region (0.30 – 1.50 A), confirming that the linear model is appropriate. Slight negative residuals at 1.40 – 1.50 A hint at the onset of saturation.



Key Numerical Findings

Metric	Value
Threshold current (motor starts spinning)	≈ 0.25 A
Slope of linear fit	1556 RPM / A
Y-intercept of linear fit	–296 RPM (extrapolated)
R ² of linear regression (0.3–1.5 A)	0.9978
Maximum observed RPM (2.00 A)	≈ 2 348 RPM
Average std dev per measurement set	5.8 RPM

6. Discussion

6.1 Linear Region (0.30 – 1.50 A)

The data confirm a strong positive linear relationship between current and RPM ($R^2 = 0.9978$). This aligns with back-EMF theory: as the supply voltage rises alongside current, the net available voltage ($V - I \cdot R$) increases, driving higher angular velocity. The slope of 1556 RPM/A reflects the combined effect of the motor's back-EMF constant k_e and armature resistance R .

6.2 Dead Zone ($I < 0.25$ A)

Below approximately 0.25 A, the electromagnetic torque is insufficient to overcome static (stiction) friction in the bearings and commutator brushes. This is the threshold current $I_{\min}$, below which the motor cannot initiate or maintain rotation.

6.3 Saturation Zone ($I > 1.60$ A)

Above 1.60 A the slope of RPM vs current decreases noticeably. Two mechanisms explain this: (1) Ohmic heating raises armature resistance $R(T) = R[1 + \alpha \Delta T]$, increasing voltage drop across the coil and reducing net torque; (2) the motor notes confirm "motor warm" observations at 1.70–1.80 A, consistent with thermal runaway of resistance.

6.4 Experimental Uncertainty

- RPM standard deviation averaged 5.8 RPM — less than 0.5% of full-scale, indicating excellent measurement precision.
- Run-to-run reproducibility was high; differences between runs were within the single-run standard deviation.
- Actual current differed from set current by ≤ 0.01 A in all cases (ammeter accuracy).
- No mechanical load was applied; results apply strictly to the free-running (no-load) condition.

7. Conclusions & Future Work

7.1 Conclusions

- Current *positively and linearly* controls RPM in the range 0.30 – 1.50 A, with a slope of 1556 RPM/A ($R^2 = 0.9978$).
- A threshold current of ≈ 0.25 A is required before the motor begins to spin.
- At currents above 1.60 A, thermal effects reduce the rate of RPM increase — the response saturates.
- All three experimental runs were highly reproducible, validating the measurement protocol.
- The empirical findings are fully consistent with back-EMF theory for DC motors.

Answer: Increasing current increases RPM linearly (for 0.3 – 1.5 A), then sub-linearly due to thermal saturation.

7.2 Limitations

- Only the no-load condition was tested; loaded motor behaviour may differ significantly.
- Temperature of the coil was not directly measured; thermal modelling was qualitative.
- A single motor unit was used — manufacturing tolerances between units were not characterised.

7.3 Recommendations for Future Work

- Repeat with a calibrated load (torque meter) to map the full torque-speed curve.
- Instrument coil temperature to quantify I^2R heating and model $R(T)$ explicitly.
- Extend current range beyond 2 A with active cooling to probe deeper saturation.
- Test multiple motor units to assess manufacturing variability.
- Implement PID closed-loop control using the empirical RPM–current relationship.

8. References & Acknowledgements

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Repository: github.com/syedjafri06193/Research-int/tree/main/Physics/Research-DC-Motor-RPM